The Physics of Transistor Evolution

and the Future of AI Chips

From FinFET to GAAFet, CFET, 2D Materials, and the Architecture of Tomorrow's AI Silicon

Deep Research Monograph — Incorporating Latest Results from IEDM 2025, Nature Electronics, imec & IRDS

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This document is compiled from primary literature and official industry roadmaps current as of Q1 2026. All equations are physically motivated and compileable in any modern LaTeX distribution with the `amsmath`, `physics`, and `siunitx` packages.

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1 Introduction: Why Physics Forces Architecture Change

For five decades, semiconductor scaling followed Dennard's rules: shrink every dimension by a factor κ, scale voltages by $1/\kappa$, and power density stays constant while speed improves by κ. The empirical consequence—Moore's Law—mapped neatly onto this physics. Both broke down simultaneously, and for the same reason: the transistor gate ceased to control the channel at sub-5 nm dimensions.

This monograph examines the *physics* of why each successive transistor architecture became necessary, the equations that govern device behaviour at each node, and how the transition from FinFET to Gate-All-Around (GAA) nanosheet to Complementary FET (CFET) and ultimately to 2D-material devices propagates through the design of present and future AI accelerators.

Central Thesis

The history of transistor scaling is a history of progressively recovering *electrostatic gate control* over the channel. Every architecture change—FinFET, GAAFet, CFET, 2D materials—solves the same underlying equation: reducing the natural length scale λ of the transistor to suppress sub-threshold leakage, drain-induced barrier lowering (DIBL), and quantum tunnelling. The implications for AI chips are architectural, not merely parametric.

2 Foundational Physics: The MOSFET Electrostatics

2.1 The Poisson Equation and Threshold Voltage

A MOSFET operates by modulating a potential barrier between source and drain. The channel electrostatics are governed by the 2-D Poisson equation:

$$\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = -\frac{\rho(x, y)}{\varepsilon_{\text{Si}}} \tag{1}$$

where ϕ is the electrostatic potential and ρ the charge density. The threshold voltage V_T is the gate voltage at which the surface potential ϕ_s at the channel mid-point equals $2\phi_F$ (twice the Fermi potential):

$$V_T = V_{FB} + 2\phi_F + \frac{Q_{\text{dep}}}{C_{\text{ox}}} \tag{2}$$

where V_{FB} is the flatband voltage, Q_{dep} the depletion charge density, and $C_{\text{ox}} = \varepsilon_{\text{ox}}/t_{\text{ox}}$ the gate oxide capacitance per unit area.

2.2 The Sub-threshold Swing and Its Fundamental Limit

When $V_{GS} < V_T$, the drain current decays exponentially:

$$I_D = I_0 \exp\left(\frac{q(V_{GS} - V_T)}{nk_BT}\right)\left[1 - \exp\left(-\frac{qV_{DS}}{k_BT}\right)\right] \tag{3}$$

where $n \geq 1$ is the ideality factor and $k_BT/q \approx 26\,\mathrm{mV}$ at room temperature. The *sub-threshold swing* S measures how many millivolts of gate voltage are needed to change drain current by one decade:

$$S = \frac{\partial V_{GS}}{\partial \log_{10} I_D} = n \cdot \frac{k_BT}{q} \ln 10 \approx 60 \cdot n \ \ [\mathrm{mV/decade}] \tag{4}$$

The *fundamental lower bound* at $T = 300\,\mathrm{K}$ is $S \geq 60\,\mathrm{mV/dec}$. A device with $n > 1$ cannot switch as sharply. For a planar MOSFET, where the gate controls only the top surface and the body is semi-infinite, $n \gg 1$ at short channel lengths. Gate-all-around architectures force $n \to 1$ by eliminating body current paths.

2.3 Short-Channel Effects and the Natural Length λ

The most critical quantity in transistor scaling is the *characteristic length* λ, which describes how far the drain electric field penetrates into the channel:

$$\lambda = \sqrt{\frac{\varepsilon_{\mathrm{Si}}}{\varepsilon_{\mathrm{ox}}} t_{\mathrm{Si}}\, t_{\mathrm{ox}}} \tag{5}$$

for a planar device with silicon body thickness t_{Si} and oxide thickness t_{ox}. Drain-Induced Barrier Lowering (DIBL) — the reduction of the channel potential barrier by the drain voltage — scales as:

$$\Delta V_T \approx -\eta_{\mathrm{DIBL}} \cdot V_{DS} \tag{6}$$

Short-channel effects (SCEs) become severe when the gate length $L_g \lesssim 5\lambda$. For acceptable SCE suppression we need $L_g/\lambda > 5$, or equivalently:

$$L_g > 5\sqrt{\frac{\varepsilon_{\mathrm{Si}}}{\varepsilon_{\mathrm{ox}}} t_{\mathrm{Si}}\, t_{\mathrm{ox}}} \tag{7}$$

This single equation explains the entire history of transistor architecture: every subsequent design reduces λ.

2.4 Quantum Tunnelling: The Absolute Wall

At gate lengths below $\sim 5\,\mathrm{nm}$, direct source-to-drain tunnelling becomes non-negligible. The Wentzel–Kramers–Brillouin (WKB) tunnelling current is:

$$J_{\text{tun}} \propto \exp\left(-\frac{2}{\hbar}\int_{x_1}^{x_2}\sqrt{2m^*\left[V(x)-E\right]}\,dx\right) \tag{8}$$

where m^* is the electron effective mass, $V(x)$ the channel potential barrier, and the integral runs from source-side turning point x_1 to drain-side x_2. For a triangular barrier of height ϕ_B and width L_g:

$$J_{\text{tun}} \propto \exp\left(-\frac{4}{3}\frac{\sqrt{2m^*\phi_B^3}}{\hbar\mathcal{E}}\right) \quad , \quad \mathcal{E} = \phi_B/L_g \tag{9}$$

This is fundamentally uncontrollable by the gate: once the barrier width L_g falls below $\sim 3\,\text{nm}$, leakage current is dominated by quantum tunnelling regardless of gate geometry. This sets the *absolute minimum channel length* for silicon-based devices.

3 FinFET: The First 3-D Transistor

3.1 Architecture and Electrostatics

The FinFET addresses the planar SCE crisis by replacing the flat silicon body with a narrow, vertical silicon *fin* of width W_{fin}. The gate wraps around *three* sides of the fin (double-gate top + sidewalls), dramatically improving electrostatic control. For a double-gate (DG) FinFET:

$$\lambda_{\text{DG}} = \sqrt{\frac{\varepsilon_{\text{Si}}}{2\varepsilon_{\text{ox}}}\frac{W_{\text{fin}}}{2}t_{\text{ox}}} \tag{10}$$

Comparing with Eq. (5): the effective body thickness is now $t_{\text{Si}} \to W_{\text{fin}}/2$ (half the fin width, because the gate attacks from both sides). For a fin width of $W_{\text{fin}} = 7\,\text{nm}$ and $t_{\text{ox}} = 1\,\text{nm}$:

$$\lambda_{\text{DG}} \approx \sqrt{\frac{11.7}{2\times 3.9}\cdot\frac{7}{2}\cdot 1} \approx 2.3\,\text{nm} \tag{11}$$

meaning SCEs are suppressed for $L_g > 5\times 2.3 \approx 12\,\text{nm}$, consistent with production FinFETs at the 5 nm node.

3.2 Drive Current and the Quantised Fin Pitch

The drive current in a FinFET scales in *discrete steps*:

$$I_D = N_{\text{fin}}\cdot I_{\text{fin}} \quad , \quad I_{\text{fin}} = \mu_{\text{eff}}C_{\text{ox}}\frac{W_{\text{eff}}}{L_g}\left(V_{GS}-V_T-\frac{V_{DS}}{2}\right)V_{DS} \tag{12}$$

where N_{fin} is the number of fins and $W_{\text{eff}} = 2H_{\text{fin}} + W_{\text{fin}}$. The *quantised* nature of N_{fin} (integers only) is a fundamental design inflexibility that GAAFet nanosheets overcome.

3.3 Why FinFET Fails Below 3 nm

1. **Fin aspect ratio:** Below 5 nm, reducing W_{fin} to control SCEs makes the fin so narrow that fin collapse during fabrication becomes probable.
2. **Bottom leakage path:** The gate does not wrap the *bottom* of the fin; a residual leakage path exists through the buried oxide.
3. **Density ceiling:** Reducing the contacted poly pitch (CPP) depopulates fins, reducing drive current per unit area.
4. **Quantum access resistance:** At sub-10 nm fin widths, quantum confinement discretises energy levels, raising source injection barriers.

4 GAAFet Nanosheet: Physics, Architecture, and the 2 nm Node

4.1 Electrostatic Advantage: The All-Around Gate

Gate-All-Around field-effect transistors (GAAFETs) extend the gate to surround *all four* sides of a channel. For a cylindrical nanowire of radius R, the characteristic length takes the elegant form:

$$\lambda_{\text{cyl}} = \frac{R}{2\sqrt{2}}\sqrt{1 + \frac{2\varepsilon_{\text{ox}}}{\varepsilon_{\text{Si}}}\frac{R}{t_{\text{ox}}}\ln\left(1 + \frac{t_{\text{ox}}}{R}\right)} \tag{13}$$

In the thin-oxide limit $t_{\text{ox}} \ll R$, this simplifies to:

$$\lambda_{\text{GAA}} \approx \sqrt{\frac{\varepsilon_{\text{Si}}}{4\varepsilon_{\text{ox}}}\frac{R^2}{t_{\text{ox}}}} \tag{14}$$

The factor of 4 in the denominator (versus 2 for DG-FinFET) reflects the additional two gating surfaces. This *halves* λ for the same body dimension, enabling shorter channels with the same SCE suppression.

For a nanosheet of width W_{NS} and thickness T_{NS}:

$$\lambda_{\text{NS}} \approx \sqrt{\frac{\varepsilon_{\text{Si}}}{\varepsilon_{\text{ox}}}\frac{T_{\text{NS}}}{4}t_{\text{ox}}} \tag{15}$$

For the TSMC N2 node: $T_{\text{NS}} \approx 5\,\text{nm}$, $t_{\text{ox, eff}} \approx 0.7\,\text{nm}$ (EOT), giving:

$$\lambda_{\text{NS, N2}} \approx \sqrt{\frac{11.7}{3.9}\cdot\frac{5}{4}\cdot 0.7} \approx 1.6\,\text{nm} \tag{16}$$

SCE suppression is guaranteed for $L_g > 5 \times 1.6 = 8\,\text{nm}$, consistent with the $\sim 12\,\text{nm}$ physical gate length at N2.

4.2 Quantum Access Resistance: The Hidden Bottleneck

A critical 2025 discovery from Kim et al. (Nature Communications Engineering, May 2025) identified *Quantum Access Resistance* (QAR) as a new performance limiter in stacked-nanosheet GAAFETs. At the constriction junction between the bulk source/drain and the narrow nanosheet channel, transverse electron energy levels change abruptly:

$$E_n^{\text{bulk}} \neq E_n^{\text{NS}} \quad \Rightarrow \quad \Delta E_n = E_n^{\text{NS}} - E_n^{\text{bulk}} > 0 \tag{17}$$

This energy mismatch creates an injection barrier analogous to a quantum point contact. Using the non-equilibrium Green's function (NEGF) formalism, the transmission probability at the constriction is:

$$\mathcal{T}(E) = \left| \frac{4k_{\text{bulk}}k_{\text{NS}}}{(k_{\text{bulk}} + k_{\text{NS}})^2} \right|^2 \exp(-2\kappa L_{\text{constr}}) \tag{18}$$

where k_{bulk}, k_{NS} are wavevectors in bulk S/D and nanosheet regions respectively, and κ is the evanescent wavevector in the constriction. The QAR degrades the on-current by 10–25% in realistic 2/2.5-D simulations. The recommended fix—a "dog-bone" shaped nanosheet extension with moderate contact depth—reduces QAR by softening the confinement gradient.

4.3 Nanosheet Width: The Drive-Current Controllability Advantage

Unlike FinFETs, where I_D is quantised, nanosheet GAAFETs allow continuous current tuning via W_{NS}. The total drive current for a 3-stack device:

$$I_{D,\text{total}} = \sum_{i=1}^{3} \mu_{\text{eff}}^{(i)} C_{\text{ox}} \frac{W_{\text{NS}}^{(i)}}{L_g} \left(V_{GS} - V_T - \frac{V_{DS}}{2} \right) V_{DS} \tag{19}$$

Each nanosheet width $W_{\text{NS}}^{(i)}$ can in principle be varied independently by multi-step etching, enabling fine-grained power/performance co-optimisation within a single standard cell.

However, a critical trade-off exists: as W_{NS} increases, lateral SCEs re-emerge because the gate can no longer fully control the *wide* channel. Kim et al. (Advanced Science, 2025) demonstrated that Morphotropic Phase Boundary (MPB) ferroelectric-antiferroelectric HZO gate dielectrics can compensate this, delivering steep sub-threshold swing ($S < 63\,\text{mV/dec}$) even for wide nanosheets.

4.4 The High-κ / Metal Gate Stack

The gate stack at N2 consists of:

1. **Interfacial SiO_2:** $\sim 0.4\,\text{nm}$, passivating Si dangling bonds
2. **HfO_2 high-κ:** $\sim 2\,\text{nm}$, $\varepsilon_r \approx 22$, giving EOT $= 2 \times (3.9/22) \approx 0.35\,\text{nm}$
3. **TiN/TiAlC/TiN work function metal (WFM):** Tunable effective work function ϕ_{EWF} from $4.47\,\text{eV}$ (n-FET) to $4.77\,\text{eV}$ (p-FET) via TiAlC thickness modulation (confirmed by Rutherford Backscattering Spectroscopy in 2024 studies)
4. **Fill metal (W or Ru):** Contacts gate to M0 interconnect

The effective oxide thickness (EOT) determines gate capacitance:

$$C_{\text{gate}} = \frac{\varepsilon_0 \varepsilon_{\text{SiO}_2}}{\text{EOT}} \approx \frac{8.85 \times 10^{-12} \times 3.9}{0.7 \times 10^{-9}} \approx 49\,\mu\text{F/cm}^2 \tag{20}$$

4.5 TSMC N2: Quantitative Performance Metrics

Table 1: TSMC N2 vs N3E: Published Performance Deltas (2025)

Metric	**N3E (FinFET)**	**N2 (GAAFet)**
Gate Architecture	FinFET (3-sided)	Nanosheet GAA (4-sided)
CPP [nm]	48	42
Fin/Sheet Pitch	25 nm fin pitch	4-sided wrap
Performance gain (iso-power)	baseline	+15%
Power reduction (iso-perf.)	baseline	−30%
Logic density gain	baseline	+20%
SRAM density gain	baseline	+10%
λ [nm]	≈ 2.8	≈ 1.6
Sub-threshold swing S	$\sim 68\,\text{mV/dec}$	$\sim 63\,\text{mV/dec}$
DIBL [mV/V]	~ 80	~ 40

5 Forksheet FET: Bridging GAAFet and CFET

The forksheet device, introduced by imec in 2017 and presented in its outer-wall variant at VLSI 2025, is a transitional architecture designed to extend the nanosheet era to the A10 node ($\sim 90\,\text{nm}$ cell height) before CFET becomes manufacturable.

In a forksheet, n- and p-type nanosheets are separated by a thin dielectric wall placed at the standard cell boundary. This eliminates the n–p spacing within the cell without the extreme aspect-ratio fabrication challenge of full CFET. The effective nanosheet width available per cell height H_{cell}:

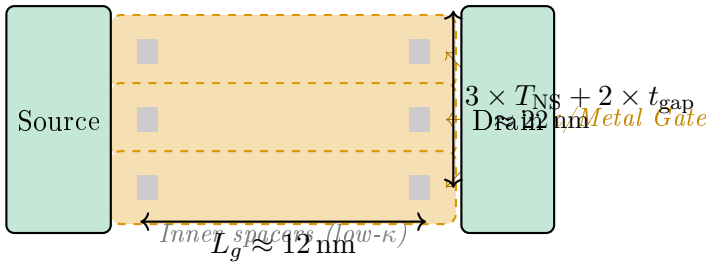

Figure 1: Cross-sectional schematic of a 3-stack GAAFet nanosheet at the TSMC N2 node. The gate metal (dashed border) completely encircles each nanosheet, providing 4-sided electrostatic control. Inner spacers (grey) are low-κ dielectrics that reduce parasitic source/drain-to-gate capacitance.

$$W_{\mathrm{eff}} = H_{\mathrm{cell}} - t_{\mathrm{wall}} - 2\,\delta_{\mathrm{margin}} \tag{21}$$

where $t_{\mathrm{wall}} \approx 15\,\mathrm{nm}$ (outer-wall design) and δ_{margin} is the lithographic placement margin.

imec VLSI 2025 simulations confirm:

- Full channel stress achievable (unlike inner-wall design)
- Connected n–p gates with no performance penalty
- Ring oscillator delay $\approx 5\%$ lower than equivalent nanosheet cell

6 Complementary FET (CFET): Stacking the Third Dimension

6.1 Architecture: Removing n–p Spacing From the Footprint

The Complementary FET stacks an n-type GAAFet directly on top of a p-type GAAFet (or vice versa) within the same horizontal footprint. This eliminates the n–p separation from standard cell height considerations for the first time in CMOS history:

$$H_{\mathrm{cell}}^{\mathrm{CFET}} = H_{\mathrm{cell}}^{\mathrm{NS}} - \Delta H_{np} \tag{22}$$

where ΔH_{np} is the horizontal n–p gap in a nanosheet cell ($\sim 20\,\mathrm{nm}$ at A10). This enables track heights to shrink from 5T to 4T and below, compressing standard cell area by $\sim$10–20%.

6.2 The Monolithic Integration Challenge

imec's preferred integration path is *monolithic* CFET, where both n- and p-FETs are fabricated in a continuous process flow:

1. Deposit alternating $Si/Si_{0.7}Ge_{0.3}$ superlattice (6–8 total layers)
2. Pattern and etch fin structure to high aspect ratio (HAR)
3. Form middle dielectric isolation (MDI) layer between n and p tiers
4. Form inner spacers for both tiers
5. Selective etch of $Si_{0.7}Ge_{0.3}$ releases Si nanosheets (critical: must release all layers without damaging MDI)
6. Deposit gate metal from all sides (common gate)
7. Form *stacked* source/drain contacts: bottom contact via backside wafer bonding + thinning

The breakthrough result announced at VLSI 2024 and confirmed in 2025: imec demonstrated the first functional monolithic CMOS CFETs at 18 nm gate length, 60 nm gate pitch, 50 nm vertical separation between n and p devices, with *top device survival rate improved from 11% to 79%* by moving bottom contact formation to the wafer backside.

6.3 Key CFET Physics: The Middle Dielectric Isolation

The MDI must electrically isolate the top-tier n-FET from the bottom-tier p-FET while being ultra-thin to minimise vertical height. The leakage through a dielectric of thickness d_{MDI} and bandgap E_g^{MDI}:

$$J_{\mathrm{MDI}} \approx A^* T^2 \exp\left(-\frac{\phi_B}{k_B T}\right) \exp\left(-d_{\mathrm{MDI}} \sqrt{\frac{8\pi^2 m^* \phi_B}{h^2}}\right) \tag{23}$$

Silicon nitride (Si_3N_4, $E_g \approx 5.3\,\mathrm{eV}$) is the current MDI material of choice, with $d_{\mathrm{MDI}} \approx$ 5–8 nm.

6.4 CFET Roadmap and AI Implications

According to imec's roadmap as of IEDM 2024:

Table 2: CFET Technology Roadmap (imec IEDM 2024 / VLSI 2025)

Node	Year	CPP	Cell Height
A14 (Nanosheet)	2028	42 nm	∼100 nm (5T)
A10 (Forksheet)	2029–2030	36 nm	90 nm (5T→4T)
A7 (CFET)	2031+	42 nm	<80 nm (4T)
A3 (CFET+2D)	2034+	TBD	<60 nm (<4T)

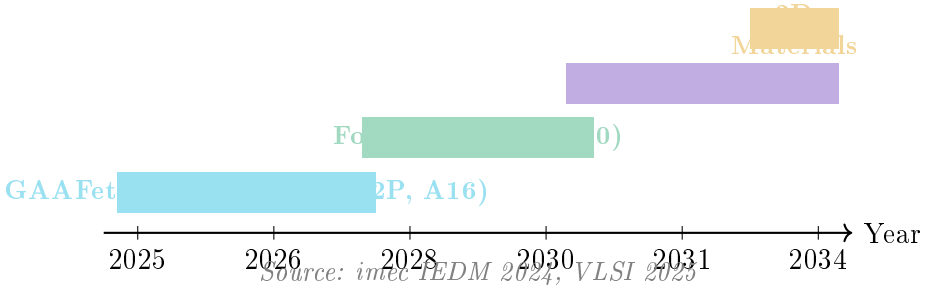

Figure 2: Transistor architecture roadmap per imec (IEDM 2024 / VLSI 2025). GAAFet nanosheets dominate through at least A10; forksheet is a bridging technology before CFET mass production around A7 (projected 2031).

7 2D Material Transistors: The Angstrom-Era Channel

7.1 Why Silicon Must Be Replaced

For CFET transistors to reach the A2 node, channel lengths must fall below 10 nm. The Poisson equation demands proportionally thin channels:

$$t_{\mathrm{Si,\,required}} < 10\,\mathrm{nm} \quad \text{for} \quad L_g < 10\,\mathrm{nm} \tag{24}$$

But silicon films thinner than ∼3 nm exhibit severe phonon scattering from surface roughness, and quantum confinement pushes subband energies to $\Delta E \sim \hbar^2\pi^2/(2m^* t_{\mathrm{Si}}^2)$, raising threshold voltage uncontrollably. The mobility degrades as:

$$\mu_{\mathrm{eff}}(t_{\mathrm{Si}}) \approx \mu_0 \exp\left(-\frac{t_0}{t_{\mathrm{Si}}}\right) \quad , \quad t_0 \approx 1.5\,\mathrm{nm} \tag{25}$$

2D transition metal dichalcogenides (TMDs) of formula MX_2 (M = Mo, W; X = S, Se, Te) are *inherently* atomically thin—a monolayer of MoS_2 is exactly ∼ 0.65 nm thick—with pristine, dangling-bond-free surfaces.

Table 3: Key 2D TMD materials for future transistor channels (IEDM 2023–2025 data)

Material	**Type**	**Bandgap [eV]**	μ_e **[cm^2/Vs]**	I_{on} **[μA/μm]**
MoS_2	n-type	1.8 (mono)	100–300	451 (IEDM 2024)
WS_2	n-type	2.0 (mono)	80–200	>300 (est.)
WSe_2	p-type	1.7 (mono)	100–200	New record (IEDM 2025)
Graphene	ambipolar	0 (zero gap)	10 000+	(no off-state)
InSe	n-type	1.3 (bulk)	287 (NIMS 2025)	TBD

7.2 Bandgap and Carrier Mobility of Key 2D Materials

The theoretical channel electrostatics for a monolayer 2D FET are optimal: since the channel is truly 2D, the characteristic length becomes:

$$\lambda_{2\text{D}} = \sqrt{\frac{\varepsilon_{\text{ch}}}{C_{\text{gate}}} t_{\text{ch}}} \quad , \quad t_{\text{ch}} \approx 0.65\,\text{nm} \tag{26}$$

For $\varepsilon_{\text{ch}} \approx 7$ (MoS_2) and $C_{\text{gate}} = 49\,\mu\text{F/cm}^2$:

$$\lambda_{\text{MoS}_2} \approx \sqrt{\frac{7 \times 8.85 \times 10^{-12}}{49 \times 10^{-2}} \times 0.65 \times 10^{-9}} \approx 0.9\,\text{nm} \tag{27}$$

This $\lambda_{2\text{D}} < 1\,\text{nm}$ enables *sub-1 nm gate lengths* while maintaining $L_g/\lambda > 5$, reaching the true Angstrom Era.

7.3 Latest Experimental Results (IEDM 2024–2025)

- **IEDM 2024, imec:** First functional stacked nanosheet FETs with monolayer MoS_2 channels deposited via layer transfer on 300 mm target wafers. $I_{\text{on}} \approx 451\,\mu\text{A}/\mu\text{m}$, $I_{\text{on}}/I_{\text{off}} > 10^9$ — both at record levels for 2D devices.

- **IEDM 2025, imec + TSMC:** Breakthrough p-type FETs with monolayer WSe_2 channels. New gate-stack integration technique using selective oxide etch on AlOx/HfO_2/SiO_2 cap; simultaneous lateral AlOx etch during contact formation reduces EOT, improving transfer characteristics. World-first fab-compatible damascene-style top contacts for 2D-FETs.

- **IEDM 2025, imec + Intel:** 300 mm manufacturable modules for n-type (WS_2 and MoS_2) and p-type (WSe_2) 2D-FETs demonstrated in same flow—a critical step for CMOS 2.0.

The CMOS 2.0 Problem

No single 2D material supports both n- and p-type operation efficiently. MoS_2 and WS_2 are optimal n-type; WSe_2 is the most promising p-type candidate. Integrating two different 2D materials in a single CFET—with atomic-level precision, on 300 mm wafers, at sub-500°C—is the central manufacturing challenge of the A3 node.

7.4 Neuromorphic 2D Synaptic Transistors

Beyond classical CMOS, MoS_2/WSe_2 heterostructures function as *artificial synapses* via charge trapping in van der Waals interfaces. The synaptic weight w (conductance) modulates as:

$$\frac{dw}{dt} = \eta_+ f_+(V_{\text{pre}}, V_{\text{post}}) - \eta_- f_-(w) \tag{28}$$

This is Spike-Timing-Dependent Plasticity (STDP) implemented in hardware, with no external memory fetch required. MoS_2/h-BN/graphene/h-BN van der Waals heterostructures have demonstrated both volatile (short-term memory, STM) and non-volatile (long-term memory, LTM) characteristics in a single device by varying gate pulse number and frequency—a direct analogue of biological synaptic consolidation.

8 Present AI Chips at the Transistor Level

8.1 NVIDIA Blackwell (B200/GB200): 208 B Transistors at 4 nm FinFET

8.1.1 Architecture Overview

NVIDIA Blackwell breaks the reticle limit by fusing two GB100 dies (∼104 B transistors each) into a single logical chip via a 10 TB/s NV-HBI die-to-die interconnect. Fabricated on TSMC N4P (an enhanced 4 nm FinFET process), Blackwell is paradoxically a *pre-GAAFet* chip—its transistors are still FinFETs. The architectural leaps are at the packaging and arithmetic level.

8.1.2 FP4 Tensor Cores: Arithmetic Efficiency at the Transistor Level

The Blackwell Tensor Core supports FP4 (4-bit floating point) matrix multiply. The arithmetic throughput scales inversely with bit-width for the same physical area:

$$\text{TOPS} \propto \frac{A_{\text{core}}}{b^2} \tag{29}$$

where b is the operand bit-width and A_{core} is the core area. Moving from FP8 ($b = 8$) to FP4 ($b = 4$) theoretically quadruples throughput (practically $\sim 2\times$ after circuit overheads):

$$\text{TOPS}_{B200,\,\text{FP4}} \approx 18\,\text{PFLOPS} \tag{30}$$

$$\text{TOPS}_{H100,\,\text{FP8}} \approx 3.9\,\text{PFLOPS} \tag{31}$$

representing a $4.6\times$ improvement, primarily from precision halving and architecture improvements.

8.1.3 Power Density and the Thermal Constraint

The B200 TDP is $\sim$1000 W for a $\sim 800\,\text{mm}^2$ die pair. Power density:

$$P_{\text{density}} = \frac{1000\,\text{W}}{800 \times 10^{-6}\,\text{m}^2} \approx 1250\,\text{W/cm}^2 \tag{32}$$

This exceeds the thermal limit for air cooling ($\lesssim 50\,\text{W/cm}^2$ at the package level) by $25\times$, mandating liquid immersion or cold-plate cooling. The GB200 NVL72 rack system draws $\sim$120 kW total.

8.2 Groq LPU: Determinism and the SRAM Architecture

8.2.1 The Von Neumann Bottleneck in Mathematical Form

The roofline model captures the GPU's fundamental constraint. Given:

- P: peak floating-point performance [FLOPS]
- B: memory bandwidth [bytes/s]
- I: arithmetic intensity of workload [FLOPs/byte]

The attainable performance is:

$$P_{\text{attain}} = \min(P,\ B \cdot I) \tag{33}$$

LLM inference (batch size = 1) has $I \approx 1$–2 FLOP/byte—a deeply memory-bandwidth-limited workload. A GPU with $P = 3.9$ PFLOPS and $B = 3.35$ TB/s is limited to:

$$P_{\text{attain}} = \min(3900,\ 3.35 \times 10^3 \times 1.5) \approx 5\,\text{TFLOPS} \tag{34}$$

— only 0.13% of peak! This is why a 3.9 PFLOP GPU generates tokens slowly at batch=1. Groq's LPU addresses this by raising B to $\sim$ 80 TB/s (on-chip SRAM) and

reducing P (no FP16 matrix math engine), optimising for the actual operating point of the workload.

8.2.2 SRAM Transistor Count: The 6T Cell

Each SRAM bit requires 6 transistors:

$$N_{\text{transistors, SRAM}} = 6 \times C_{\text{SRAM}} \tag{35}$$

where C_{SRAM} is capacity in bits. For the Groq HS-1 chip (estimated ∼220 MB SRAM on-chip):

$$N_{\text{transistors, SRAM}} = 6 \times 220 \times 10^6 \times 8 \approx 10.6 \times 10^9 \approx 10 \text{ billion} \tag{36}$$

This dominates the transistor budget, explaining why Groq's chip does not need TSMC's cutting-edge 4/3 nm FinFET: the SRAM bit-cell area at GF 14 nm is larger, but the chip doesn't need transistor density—it needs SRAM bandwidth. At 14 nm (compared to 4 nm), each transistor has:

$$\frac{R_{\text{on, 14nm}}}{R_{\text{on, 4nm}}} \approx \frac{1}{(14/4)^{0.7}} \approx 2.1 \tag{37}$$

(higher on-resistance), but this is irrelevant for SRAM sense amplifiers operating at ~ 1 GHz – the bandwidth is determined by the number of sense amplifiers firing in parallel.

8.2.3 Determinism: Zero Idle Transistors

In a GPU, the CUDA scheduler dynamically assigns warps to streaming multiprocessors. At any instant, a fraction η_{util} of arithmetic units are active:

$$P_{\text{effective}} = \eta_{\text{util}} \cdot P_{\text{peak}} \quad , \quad \eta_{\text{util}} < 1 \tag{38}$$

For memory-bound LLM inference, $\eta_{\text{util}} \approx 0.1$–$0.3$. Groq's compiler statically schedules every instruction:

$$\eta_{\text{util}}^{\text{LPU}} = 1 - \varepsilon_{\text{dead}} \approx 0.99 \tag{39}$$

where $\varepsilon_{\text{dead}}$ is a tiny compiler-enforced NOP budget. This is the physical reason Groq achieves > 1600 tokens/second where GPUs achieve 10–30.

9 Beyond Groq and NVIDIA: Future AI Hardware Paradigms

9.1 Photonic AI Accelerators and Co-Packaged Optics

9.1.1 The Interconnect Energy Crisis

By 2024, ∼30% of an AI cluster's total energy was consumed by data movement between chips. Copper electrical interconnects at 200 Gb/s per lane require:

$$E_{\text{copper}} \approx 5\text{–}20 \text{ pJ/bit} \tag{40}$$

Silicon photonics replaces copper with light. The fundamental energy for a photonic bit transmission:

$$E_{\text{photon}} = \frac{h\nu}{\eta_{\text{det}}\,\eta_{\text{mod}}} \tag{41}$$

where $h\nu \approx 0.8\,\text{eV}$ (1550 nm C-band), and η values are detector/modulator efficiencies. State-of-art Co-Packaged Optics (CPO) achieves:

$$E_{\text{CPO}} \approx 5\,\text{pJ/bit} \tag{42}$$

(confirmed by NVIDIA's IEDM 2025 presentation; Spectrum-X Photonics delivers 5× power reduction per 1.6 Tb/s port versus pluggable copper).

9.1.2 NVIDIA's Photonic Roadmap (2026)

NVIDIA Quantum-X Photonics InfiniBand switches (early 2026):

- 115 Tb/s total throughput, 144 ports at 800 Gb/s
- CPO silicon photonics engine co-packaged with switch ASIC
- 10× network resiliency improvement vs pluggable

NVIDIA Spectrum-X Photonics Ethernet (H2 2026):

- SN6800: 409.6 Tb/s, 512 ports at 800 Gb/s
- Spectrum-6 ASIC + fully integrated 512-lane 200G-capable architecture

The photonic modulator physics: a Mach-Zehnder Interferometer (MZI) based on the electro-optic effect in silicon:

$$\Delta n_{\text{eff}}(\Delta N_e, \Delta N_h) = -\frac{e^2\lambda^2}{8\pi^2 c^2 \varepsilon_0 n}\left(\frac{\Delta N_e}{m_e^*} + \frac{\Delta N_h}{m_h^*}\right) \tag{43}$$

The phase shift $\Delta\phi = 2\pi\Delta n_{\text{eff}} L_{\text{arm}}/\lambda$ enables switching between constructive and destructive interference.

9.2 Neuromorphic AI: In-Memory Compute at the Transistor Level

9.2.1 The Memristor: Physics of Analogue Resistance

A memristor stores synaptic weight as a *continuous resistance state*. The resistance dynamics of a filamentary oxide memristor (e.g., TaO_x, WO_x, HfO_x) are governed by:

$$\frac{dw}{dt} = \mu_v \frac{R_{\text{on}}}{D^2} \cdot I(t) \cdot f(w) \tag{44}$$

where $w \in [0, 1]$ is the normalised filament length, D is the oxide thickness, μ_v is the ionic drift mobility, and $f(w)$ is a window function enforcing boundary conditions. The i–v characteristic:

$$v(t) = [R_{\text{on}}\, w(t) + R_{\text{off}}\, (1 - w(t))] \cdot i(t) \tag{45}$$

This non-linear, history-dependent resistance directly encodes a synaptic weight without a separate memory access cycle—eliminating the Von Neumann bottleneck at the device level.

9.2.2 Crossbar Arrays: Parallel Matrix-Vector Multiplication

In a crossbar array of $N \times M$ memristors, vector-matrix multiplication is performed in analogue, in one clock cycle, by Kirchhoff's current law:

$$I_j = \sum_{i=1}^{N} G_{ij} V_i \quad \Rightarrow \quad \mathbf{I} = \mathbf{G}\,\mathbf{V} \tag{46}$$

where $G_{ij} = 1/R_{ij}$ encodes weight matrix $\mathbf{W}$ and $\mathbf{V}$ is the input voltage vector. The energy per operation:

$$E_{\text{MAC}} = \frac{V^2}{\bar{R}} \cdot \tau_{\text{pulse}} \sim 1\,\text{fJ} \tag{47}$$

versus $\sim$ 10 fJ for a digital MAC in 4 nm CMOS— a 10× energy advantage per multiply-accumulate.

9.2.3 Intel's First LLM on Neuromorphic Hardware (April 2025)

Eshraghian et al. (ICLR 2025) demonstrated the first LLM adapted to run on Intel Loihi 2—a spiking neural network (SNN) implementation. Key result: *same accuracy as GPU-based LLM at half the energy.* This is the proof-of-concept that neural-architecture–matched hardware (SNN $\approx$ Transformer attention, under certain re-formulations) can deliver radical efficiency improvements.

9.3 The Rubin Era: NVIDIA on 2 nm GAAFet (2026–2027)

NVIDIA's Rubin Ultra GPU will be among the first AI accelerators on TSMC's N2 GAAFet process. The implications at the transistor level:

$$\frac{P_{\text{dynamic, Rubin}}}{P_{\text{dynamic, Blackwell}}} \approx \left(\frac{V_{DD,N2}}{V_{DD,N4P}}\right)^2 \cdot \frac{f_{N2}}{f_{N4P}} \approx (0.85)^2 \times 1.15 \approx 0.83 \tag{48}$$

A ∼17% dynamic power reduction at the same frequency, plus ∼15% higher performance at iso-power—consistent with N2's published deltas. Combined with continued FP4 / FP2 precision scaling and improved MoE (Mixture-of-Experts) routing silicon, Rubin Ultra is projected to deliver $\sim 2\times$ the inference throughput of Blackwell in the same rack power envelope.

9.4 Heterogeneous Chiplet AI: The Architecture of 2030

The most important structural shift in AI hardware is the transition from monolithic dies to *heterogeneous chiplet systems-in-package.* The optimal system partitions function by process technology:

Table 4: Projected Chiplet AI System-in-Package: Technology Partition

Chiplet	**Function**	**Process**	**Why**
Compute Die A	Dense FP4/FP8 training	TSMC A14 GAAFet	Max transistor density
Compute Die B	Deterministic inference	GF / TSMC N3	SRAM-optimised, LPU-style
Memory Die	HBM4 / SRAM cache	HBM fab	Bandwidth + capacity
Photonic I/O Die	400+ Tb/s optical I/O	TSMC COUPE	Silicon photonics
Neuro Die	Memristive SNN cores	Specialty fab	<1 fJ/MAC

All dies are integrated on a silicon interposer (CoWoS-L or equivalent) or a glass interposer (for larger substrates), with advanced hybrid bonding achieving $< 1\,\mu$m bump pitch and >1 TB/s die-to-die bandwidth.

10 Backside Power Delivery: The Hidden Revolution

Traditional chips route power and signal on the same side (frontside). The power delivery network (PDN) must distribute current from package bumps to individual transistors, with resistive losses:

$$\Delta V_{\mathrm{IR}} = I_{\mathrm{supply}} \cdot R_{\mathrm{PDN}} \tag{49}$$

At N2 densities, R_{PDN} is dominated by narrow metal lines in M0–M2. Backside Power Delivery Networks (BSPDN) route power rails through the *back of the wafer* via nano-scale through-silicon vias (nTSVs):

$$R_{\mathrm{BSPDN}} = \frac{\rho_{\mathrm{W}} \cdot L_{\mathrm{nTSV}}}{A_{\mathrm{nTSV}}} \approx \frac{53\,\mathrm{n\Omega\,m} \times 200\,\mathrm{nm}}{\pi(10\,\mathrm{nm})^2} \approx 34\,\Omega \tag{50}$$

per nTSV, but with millions in parallel per die the total ΔV_{IR} is reduced by ∼15–30 mV, enabling V_{DD} reduction and performance improvement.

TSMC's Super Power Rail (SPR) in A16 directly contacts transistor source/drain from the backside, representing the most aggressive BSPDN implementation to date.

11 Energy as the Governing Constraint: The Path Forward

The total energy to generate one LLM inference token can be decomposed as:

$$E_{\mathrm{token}} = E_{\mathrm{compute}} + E_{\mathrm{DRAM}} + E_{\mathrm{interconnect}} \tag{51}$$

$$E_{\mathrm{compute}} = N_{\mathrm{params}} \cdot 2 \cdot E_{\mathrm{MAC}} \tag{52}$$

$$E_{\mathrm{DRAM}} = N_{\mathrm{params}} \cdot B_{\mathrm{weight}} \cdot e_{\mathrm{DRAM}} \tag{53}$$

where $e_{\mathrm{DRAM}} \approx 10\,\mathrm{pJ/B}$ (HBM3e) and B_{weight} is bytes per parameter (2 for FP16, 0.5 for FP4). Each technology transition shifts this balance:

- **GAAFet (N2):** Reduces E_{compute} by ∼30% (lower V_{DD})
- **FP4:** Halves B_{weight} and E_{compute}
- **CPO (silicon photonics):** Reduces $E_{\mathrm{interconnect}}$ by 5×
- **Memristive crossbars:** Reduces E_{compute} by 10× and eliminates E_{DRAM} (in-memory compute)
- **CFET (A7, 2031+):** Reduces E_{compute} by further 40% via density doubling and continued V_{DD} scaling
- **2D materials (A3+):** Enables sub-thermal-limit switching via ferroelectric FETs ($S < 60\,\mathrm{mV/dec}$), potentially reducing E_{compute} by another 2–5×

The combined roadmap suggests AI inference energy per token could fall by $\sim 100\times$ between 2025 and 2035.

12 Synthesis: What the Transistor-Level Physics Means for Future AI Chip Design

The Deep Principle

Every transistor architecture transition since 2000 has been driven by one equation: reducing λ while maintaining current drive. The AI workload evolution—from dense matrix multiply (GPUs) to sparse attention (Transformers) to continuous learning (neuromorphic)—is converging with the transistor roadmap in a way that makes *architecture specialisation* more valuable than raw transistor count.

The architectural evolution of AI chips maps onto the transistor roadmap as follows:

2025–2026: GAAFet Era (N2/A16) NVIDIA Rubin, AMD Instinct MI450. GAAFet gives $\sim$30% power reduction, enabling $>$1 MW AI training clusters without proportional energy cost growth. Groq LPU-style architectures (whether via NVIDIA's acquisition or competitors) scale to smaller nodes with massive SRAM bandwidth.

2027–2030: Forksheet / Early CFET Era (A14/A10) Heterogeneous chiplet packages become standard. The first *photonic AI systems* with CPO at $>$100 Tb/s interconnect appear. Memristive in-memory compute chips (ReRAM/MRAM crossbars) begin to appear as inference accelerator tiles, delivering < 1 fJ/MAC.

2031–2035: CFET Era (A7/A3) Cell heights shrink below 80 nm. AI chip transistor density effectively doubles relative to GAAFet nanosheets. Neuromorphic tiles with 2D-material synaptic transistors ($> 10^9$ synapses on a single die) achieve brain-comparable density. The LLM inference energy cost falls below 1 nJ/token.

2036+: 2D Material CFET (A2 and beyond) Silicon channels replaced by MoS_2 (n) and WSe_2 (p) in CFET geometry. $\lambda < 1$ nm enables sub-nm gate lengths. Ferroelectric gate dielectrics ($S < 60$ mV/dec) break the Boltzmann tyranny. AI inference hardware achieves brain-comparable energy efficiency: ~ 1 fJ/synapticoperation.

13 Conclusions

The arc of transistor evolution is fundamentally a story about electrostatics. Planar MOSFETs lost gate control when $L_g/\lambda < 5$. FinFETs recovered it by shrinking the effective body to $W_{\text{fin}}/2$. GAAFet nanosheets doubled the improvement by closing all four sides of the channel. CFET will double density again without requiring further lateral scaling. And 2D materials will ultimately extend this roadmap to the true Angstrom Era by providing atomically thin channels with intrinsically pristine surfaces.

For AI chips, these transitions are not merely parametric. Each one enables a qualitatively different architecture:

- GAAFet enables continued scaling of GPU-style dense matrix compute.
- SRAM-centric, deterministic LPU architectures (Groq-style) become manufacturable at higher density on older nodes precisely because their advantage is bandwidth, not transistor count.
- Silicon photonics resolves the interconnect energy crisis that would otherwise cap AI cluster scaling.
- Memristive neuromorphic hardware eliminates the memory wall at the device level.
- 2D material CFETs enable brain-scale integration of synaptic transistors.

The AI chip of 2035 will not be a faster version of today's GPU. It will be a heterogeneous, photonically-interconnected, memristively-augmented system-in-package, operating on principles—deterministic dataflow, analogue in-memory compute, spiking neural computation—that are architecturally orthogonal to CUDA. The physics of the transistor is not a ceiling; it is a map.

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13 Key Equations Reference

Table 5: Summary of Key Physical Equations

Equation	**Physical Meaning**	**Section**
Eq. (5)	Planar MOSFET characteristic length	§2
Eq. (14)	GAA characteristic length	§4
Eq. (15)	Nanosheet GAA λ	§4
Eq. (26)	2D material λ	§7
Eq. (8)	WKB quantum tunnelling	§2
Eq. (33)	GPU roofline model	§8
Eq. (46)	Memristive crossbar MAC	§9
Eq. (51)	AI token energy budget	§11